Unit: Unit 6: Systems of Linear Equations and Inequalities **Name:** ______________

Topic: Applications of Systems (Word Problems) **Date:** __________

Challenge (Habanero)

Q1. A hiker is planning a trip to climb two mountains. The first mountain is $3 \cdot x$ miles away, and the second mountain is $2 \cdot x + 5$ miles away. If the hiker has 24 miles of hiking capacity, which system of equations represents the situation?

(A) $3x + (2x + 5) = 24$
(B) $3x - (2x + 5) = 24$
(C) $3x + (2x - 5) = 24$
(D) $3x - (2x - 5) = 24$
(E) $3x + 2x = 24$

Q2. A park ranger is tracking the population of rabbits and foxes in a forest. The rabbit population is growing at a rate of $2 \cdot t$ rabbits per year, and the fox population is growing at a rate of $t+5$ foxes per year. If the initial rabbit population is 10 and the initial fox population is 5, which system of equations represents the situation after t years?

(A) $R = 10 + 2t$, $F = 5 + t + 5$
(B) $R = 10 + t$, $F = 5 + 2t$
(C) $R = 10 - 2t$, $F = 5 - t - 5$
(D) $R = 10 - t$, $F = 5 - 2t$
(E) $R = 10 + t$, $F = 5 + t$

Q3. A group of friends want to go on a road trip that is 240 miles long. If their car gets 30 miles per gallon, and gas costs

$$2.50$$

per gallon, which system of equations represents the cost C and the number of gallons g of gas consumed?

(A) $C = 2.5g$, $240 = 30g$
(B) $C = 2.5g$, $240 = g$
(C) $C = 2.5 + g$, $240 = 30g$
(D) $C = 2.5 - g$, $240 = 30g$
(E) $C = 2.5g$, $240 = 30 - g$

Q4. A farmer is planning to plant two crops, corn and wheat, on his land. The profit from corn is

$$4$$

per acre, and the profit from wheat is

$$3$$

per acre. If the farmer has 100 acres of land and wants to make a total profit of

$$320$$

, which system of equations represents the situation?

(A) $4c + 3w = 320$, $c + w = 100$
(B) $4c - 3w = 320$, $c - w = 100$
(C) $4c + 3w = 320$, $c - w = 100$
(D) $4c - 3w = 320$, $c + w = 100$
(E) $4c + 3w = 100$, $c + w = 320$

Q5. A group of hikers are planning a trip to a mountain. They will be hiking for t hours at a rate of 2 miles per hour. If they start at an elevation of 1000 feet and climb at a rate of 500 feet per hour, which system of equations represents their distance d and elevation e after t hours?

(A) $d = 2t$, $e = 1000 + 500t$
(B) $d = 2t$, $e = 1000 - 500t$
(C) $d = 2t$, $e = 1000 + t$
(D) $d = t$, $e = 1000 + 500t$
(E) $d = t$, $e = 1000 - 500t$

Q6. A park is home to a population of deer and a population of wolves. The deer population is growing at a rate of 5 deer per year, and the wolf population is growing at a rate of 2 wolves per year. If the initial deer population is 50 and the initial wolf population is 10, which system of equations represents the situation after t years?

(A) $D = 50 + 5t$, $W = 10 + 2t$
(B) $D = 50 - 5t$, $W = 10 - 2t$
(C) $D = 50 + 2t$, $W = 10 + 5t$
(D) $D = 50 - 2t$, $W = 10 - 5t$
(E) $D = 50 + t$, $W = 10 + t$

Q7. A group of friends want to rent a car for a road trip. The rental company charges a base fee of

$$40$$

plus an additional

$$0.25$$

per mile. If the friends want to spend no more than

$$120$$

, which system of equations represents the cost C and the number of miles m driven?

(A) $C = 40 + 0.25m$, $Cleq120$
(B) $C = 40 - 0.25m$, $Cleq120$
(C) $C = 40 + m$, $Cleq120$
(D) $C = 40 - m$, $Cleq120$
(E) $C = 0.25m$, $Cleq120$

Q8. A farmer is planning to harvest two crops, apples and pears. The profit from apples is

$$1.50$$

per bushel, and the profit from pears is

$$2$$

per bushel. If the farmer has 100 bushels of crops to harvest and wants to make a total profit of

$$150$$

, which system of equations represents the situation?

(A) $1.5a + 2p = 150$, $a + p = 100$
(B) $1.5a - 2p = 150$, $a - p = 100$
(C) $1.5a + 2p = 150$, $a - p = 100$
(D) $1.5a - 2p = 150$, $a + p = 100$
(E) $1.5a + 2p = 100$, $a + p = 150$

Open Ended Questions (FRQ)

Q9. A group of friends want to go on a camping trip to a lake. The distance to the lake is 15 miles, and they will be hiking at a rate of 2.5 miles per hour. If they start at an elevation of 500 feet and climb at a rate of 200 feet per hour, find their elevation after 3 hours. Show your work and explain your answer.

Q10. A park is home to a population of rabbits and a population of hawks. The rabbit population is growing at a rate of 8 rabbits per year, and the hawk population is growing at a rate of 3 hawks per year. If the initial rabbit population is 80 and the initial hawk population is 20, find the populations after 5 years. Show your work and explain your answer.

Chef's Tips

- Read each question carefully and choose the correct answer.
- Show your work and explain your answers for free-response questions.
- Use a calculator only when necessary and show your work.